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Lighting Res. Technol. 2019; **0**: 1–21

The Society of Light and Lighting

Influence of indoor temperature and daylight illuminance on visual perception

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Received 28 December 2018; Revised 24 May 2019; Accepted 4 June 2019

This paper details the results of an experimental study investigating the combined effect of indoor temperature (19 °C, 23 °C and 27 °C) and horizontal illuminances (~140 lx, ~610 lx and ~1440 lx) on visual perception, with daylight as the only source of light. The goal is to evaluate the cross-modal effect of temperature and the unimodal effect of daylight illuminances on the visual perception of daylight. A 3×3 mixed-design experiment was conducted in an office-like test room with 84 participants. Results show that cross-modal effects of indoor temperature on visual perception occurred: Temperature influenced the perceived warmth of daylight, and temperature interacted with daylight illuminance as the high illuminance was considered more pleasant, was better evaluated and was more often chosen than the low one only in a thermally comfortable condition. In terms of unimodal effects, differences in visual comfort evaluations were substantial only between the low and the other two illuminances, suggesting that visual comfort did not increase with illuminance above an already comfortable threshold. Moreover, perceived brightness under the high daylight illuminance was lower than that reported in past studies under a comparable electric light illuminance, leading to the hypothesis that high illuminances might be more tolerated in a naturally-lit environment compared to a space illuminated with electric light. Findings advance the understanding of occupant perceptions of daylit environments and suggest that, together with visual factors, thermal conditions should be considered in visual perception investigations.

1. Introduction

The perception of an indoor visual environment is strongly affected by light and its characteristics. Light quantity, distribution, colour and risk of discomfort glare all have implications for human visual perception. The horizontal illuminance on the working plane, used to measure light quantity, is one of the most widespread parameters considered in building design[1]. With a focus on electric light, many studies have investigated the relationship between illuminances and various aspects of human visual perception, from the preferred illuminance and perceived brightness to the switch-on probability of electric light[2–8].

In addition to electric light, it is essential to understand how building occupants visually perceive a daylit space, considering the importance of daylight for occupant comfort, well-being and health[1,9–13]. With a focus on

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10.1177/1477153519859609

daylight quantity, knowing how occupants perceive indoor illuminances can influence building design and operation choices (e.g., openings, glazing properties, shading devices, building shape and surroundings). However, current knowledge on how people respond on a psychological level to daylight is still limited, and further investigations are needed[10]. In particular, large differences among people regarding preferred and tolerated daylight levels have been reported in previous studies[14–20]. These results may be due to individual differences but also to the type of tasks and to variations of other visual aspects. To the authors' knowledge, a controlled experiment investigating the effects of variations of daylight illuminances on visual perception, with daylight levels maintained below glare thresholds, is missing from the literature. Such an investigation would advance the understanding of the effects of daylight illuminances on visual perception, including, but not limited to, perceived brightness and preference.

Further investigations with the use of controlled daylight levels would also shed light on other open questions, such as the impact of non-visual-related parameters on visual perception (e.g., tasks or characteristics of the individuals[1,21]). Considering that people experience indoor environmental stimuli simultaneously (i.e., visual, thermal, acoustic and air quality), non-visual indoor factors could also be part of the parameters affecting visual perception. Given the more established cross-modal influence of visual stimuli on thermal responses of people[22], it is plausible that thermal conditions could affect occupant perception of the visual environment. Regardless, researchers have questioned the unidirectionality of this visual–thermal interaction because temperature is not a fundamental dimension of light evaluation[23]. The question is still open as only a few studies have investigated the cross-modal effect of temperature on visual perception considering illuminance levels or daylight[24–29]. The studies examining the effect of temperature on visual perception at different illuminances[24,25,29] employed electric light as the only source of light. Other studies used daylight[26,27,30] but did not strictly control or vary illuminances within a comfortable range (i.e., daylight illuminance was only measured[26,30] or high light intensities were used to induce glare[27]). As a consequence, no study exists on the effect of indoor temperature on the visual perception of daylight at various non-glaring illuminances.

To fill these research gaps, this study describes a controlled experiment investigating the combined effect of illuminance and temperature levels on visual perception with the use of daylight. Cross-modal effects of temperature on visual perception as well as temperature interactions with illuminances are considered with the aim to answer the following questions:

– Are visual evaluations influenced by indoor temperature? (cross-modal main effect)
– If there is a significant effect of temperature on visual evaluations, does such an influence depend on the illuminance? (cross-modal interaction effect)

Furthermore, the unimodal influence of daylight illuminances on visual perception is explored to highlight possible differences in comparison to studies on electric light, and to advance the understanding of occupant perception of daylit environments.

2. Method

2.1. Participants

A total of 84 participants took part in this study, of which 42 were women and 42 men. Gender balance was also respected within each of the three temperature levels investigated, experienced by 28 participants each. Participants were volunteers and were pre-selected from the university community in the

Table 1 Characteristics of participants according to gender and temperature level (mean ± SD, whenever applicable).

Temperature	19 °C		23 °C		27 °C	
Gender	Women	Men	Women	Men	Women	Men
Age [years]	19.9 ± 1.5	18.8 ± 1.1	19.5 ± 1.2	19.0 ± 0.9	19.3 ± 1.5	18.7 ± 0.9
Weight [kg]	59.3 ± 8.7	69.2 ± 6.6	56.5 ± 5.9	68.0 ± 5.8	56.6 ± 7.5	68.5 ± 8.6
Height [m]	1.6 ± 0.1	1.8 ± 0.1	1.6 ± 0.1	1.7 ± 0.1	1.6 ± 0.1	1.8 ± 0.1
BMI [kg/m^2]	21.6 ± 2.8	21.3 ± 1.6	20.4 ± 1.6	21.6 ± 2.3	21.3 ± 2.8	21.0 ± 2.0

Lausanne region (Switzerland) based on the following criteria: age 18–30 years old, no abuse of alcohol or use of drugs, full colour vision, generally healthy, French speaking – mother tongue or C2 level, body mass index between 18 kg/m^2 and 25 kg/m^2 and no visual or motor abnormalities. People aware of the goal of the experiment or in close relationship with the topic were excluded to avoid response bias. All participants had normal or correct-to-normal vision. Table 1 lists the characteristics of participants according to the gender and within the temperature level investigated.

Participants provided written consent prior to the start of the experiment and were paid for their participation. This study was approved by the Cantonal Ethics Committee of the Canton Vaud, Switzerland (CER-VD, ref. no. 2016-01115) and adhered to the principles of the Declaration of Helsinki.

2.2. Experimental setup

Experiments were performed in an office-like test room (Figure 1), with the indoor temperature regulated with a radiant system installed on all the opaque surfaces. The room presented two openings (on the South and North façades, respectively) with coated double glazing (Pilkington Optiwhite^{TM}, low iron extra clear float glass with a visual transmittance equal to 75%), modifiable with the application of colour neutral foil-filters. Two facing desks for participants were positioned close to the North façade to avoid any glare risk or direct sun on the horizontal

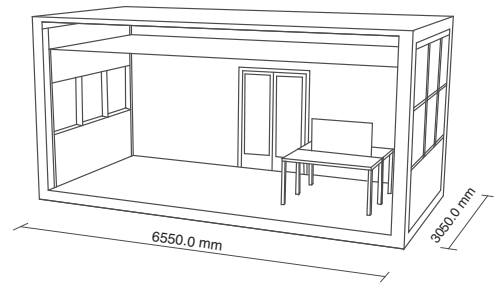


Figure 1 Experimental test room scheme with the two desks of participants close to the north façade.

surface of the desks. It must be noted that, despite the lack of glare, sunlight was always present in the room through the south window as experiments were always performed in sunny days. Although sunlight penetration varied across experimental sessions (as they were performed over a long time span), sunrays never reached the location of the two desks of participants. A translucent white panel was placed vertically between the two desks to block the view of one participant to the other. The colour-scheme of the test room was neutral: grey floor (reflectance 27%), white walls, ceiling and door (reflectance 81–84%) and grey furniture (reflectance 48–50%). The black-out curtain used in the experimental phases without daylight was white on the inside of the room (reflectance 60%). The test room was equipped with monitoring instrumentation, recording indoor environmental factors at a one-minute sampling rate. Air and globe temperatures were measured at a distance of 50 cm



Figure 2 Internal view of the test room using glazing-filter combinations with (a) 7% visible transmittance (low illuminance), (b) 30% visible transmittance (medium illuminance) and (c) 75% transmittance (high illuminance).

from each participant and at four heights (0.1 m, 0.6 m, 1.1 m and 1.6 m), corresponding to ankle, body, head (sitting person) and head (standing person) levels according to the EN ISO 7726 standard[31]. A single value for operative temperature (from now on referred simply as 'temperature') was calculated from the air and globe temperature measurements (through the calculation of the mean radiant temperature with the air velocity and the globe diameter values[32]), averaging the results at each height. Horizontal illuminance was recorded at two points symmetrically distributed on each desk. Also in this case, an average of the two measurements was used in the analysis. For a detailed description of the test room and of the measurement equipment, refer to Chinazzo *et al.*[28]

2.3. Experimental design

A 3 × 3 full factorial mixed-design was used in the experiment to investigate the combined effect of daylight illuminance and indoor temperature on visual perception. The three levels of daylight illuminance were on average 140 lx 'low', 610 lx 'medium' and 1440 lx 'high', generated with a normal double-glazing and optional neutral foils attached to an acrylic panel. The total visible transmittance of the three glazing-foil-acrylic combinations was equal to 7% (low), 30% (medium) and 75% (high, no foil; Figure 2). The three levels of indoor temperature were 19 °C, 23 °C and 27 °C, chosen to have a comfortable thermal environment (23 °C), and two slightly uncomfortable thermal conditions compared to temperature ranges usually found in buildings, namely slightly cool (19 °C) and slightly warm (27 °C). The daylight illuminance was manipulated within participants in a counterbalanced order across temperature levels (i.e., a within-subjects factor), and the indoor temperature varied between participants in a randomised order across days (i.e., a between-subjects factor). As a result, each participant experienced all three daylight levels in a row, but only one temperature level. To maximise the experimental time as daylight was the only source of light (except in the pre- and post-phases where electric light was used), experiments were planned in both mornings and afternoons starting between 8:30 and 9:00, and between 12:30 and 13:00, respectively.

2.4. Measures

Visual perception evaluations were recorded with a subjective questionnaire presented on a tablet at two different moments in the experiment. Questions were asked first three times, at the end of each 'illuminance session' while participants were still exposed to a daylight illuminance (DE = During Exposure), and then an additional time at the end of the entire experiment, during the debriefing phase, when participants had experienced all the three illuminances, and were

Table 2 Visual perception questions in the During Exposure (DE) phase (translated from French).

	Question	Response scale
Light quantity (brightness perception) evaluation	**DE1. Visual sensation:** How do you perceive the quantity of light on your desk?	Very low (1), low (2), slightly low (3), adequate (4), slightly high (5), high (6), very high (7)
	DE2. Visual comfort: How do you find the quantity of light on your desk?	Very comfortable (5), comfortable (4), slightly uncomfortable (3), uncomfortable (2), very uncomfortable (1)
	DE3. Visual preference: In this moment, would you like to have a light on your desk?	Much brighter (1), brighter (2), slightly brighter (3), no change (4), slightly dimmer (5), dimmer (6), much dimmer (7)
	DE4. Overall visual environment comfort: How do you find the general visual environment at the room?	Very comfortable (5), comfortable (4), slightly uncomfortable (3), uncomfortable (2), very uncomfortable (1)
General lighting evaluation	**DE5. Lighting cold–warmth association:** Please specify how do you perceive the appearance of the indoor light environment at this moment:	Cold (1)–Warm (7)
	DE6. Lighting comfort: Please specify how do you perceive the appearance of the indoor light environment at this moment:	Uncomfortable (1)–Comfortable (7)
	DE7. Lighting pleasantness: Please specify how do you perceive the appearance of the indoor light environment at this moment:	Unpleasant (1)–Pleasant (7)

Table 3 Visual perception questions in the After Exposure (AE) phase (translated from French).

Question	Response item
AE1. Lighting evaluation: Imagine that you are in the room to work. Assign a grade to each of the three conditions of the experiment from 1 to 10, with 1 being not at all pleasant and 10 very pleasant.	First condition: ——— Second condition: ——— Third condition: ———
AE2. Lighting rank: Rank the three conditions of the experiment according to your preference.	First condition: ——— Second condition: ——— Third condition: ———

exposed to electric lighting (AE = After Exposure). Tables 2 and 3 describe the questions with reference to the DE and AE phases, indicating them with an identification code for ease of understanding in the Results section. As indicated in the tables, the DE questions were answered on a five-point or seven-point response scale according to commonly used scales in the literature, while the AE1 question was answered on a 10-point scale. The first three DE questions (visual sensation, comfort and preference) referred to the light quantity (i.e., brightness perception) at the desk level; DE4 relates to the overall visual environment comfort; the last three DE questions relate to the general lighting evaluation (lighting cold-warmth association, lighting comfort and lighting pleasantness). The scale of questions DE2 and DE4 (on comfort) was derived from the standards used for the subjective evaluation of the thermal environment[33], which suggests using a four- or five-point scale from 'comfortable' to 'very uncomfortable' or 'extremely uncomfortable'. We wanted to be consistent with the standardised scale (as thermal comfort was also evaluated during

the experiment[30]), but we also wanted to estimate a positive effect of the light environment. Hence, we added the response 'very comfortable' to the suggested four-point scale (excluding the response 'extremely uncomfortable'), as also reported in other studies evaluating different aspects of the indoor environment[34] or the interaction effects between visual and thermal factors with the use of electric light[24]. During the experiment, each illuminance phase was introduced as the 'first', 'second' or 'third' condition, and participants were warned that there will be questions on the comparison of the three conditions at the end of the experiment.

In the following analysis, for conciseness, the 'lighting comfort' question (DE6) will not be discussed considering the very similar results in terms of responses and of statistical analysis with the 'visual comfort' question (DE2; $r_s = 0.7$, $p < 0.001$ after Spearman's rank-order correlation test between the DE2 and DE6 questions). Similar responses on two different but similar questions can be considered as an internal validation of the questionnaire, meaning that participants were consistent with their evaluation of the indoor environment.

2.5. Procedure

Each experimental session, involving two participants at a time, lasted approximately three hours. It consisted of a 45-minute pre-test phase, 3×30-minute daylight illuminance exposures (preceded by a 10-minute break to change filters) and a 15-minute debriefing phase (Figure 3). During the pre-test phase, used for thermal and metabolic adaptation, participants signed the consent form, filled in a background questionnaire and were trained on the paper-based performance test that they had to perform during the experiment in each daylight exposure. This performance test consisted of three tasks evaluating the distributed visual attention (Tsai Partington), the sustained vigilance (d2 test) and the logical reasoning (Baddeley test). All tests were performed on white A4 paper with black text. Results of these tasks are not discussed in this paper but can be found in Chinazzo *et al.*[35]. A single-blind procedure was used in the description of the experiment, without disclosing the real purpose of the test to avoid biasing participants' responses. A blackout curtain was used on both openings to avoid daylight entering the room in this phase, during which only electric light was used (at 400 lux at the desk level, 3800 K, fluorescent lighting). A relaxation phase of 10 minutes preceded each daylight illuminance phase, in which participants were asked to relax by listening to a song of their choice, blindfolded. This time was used to change the neutral filters on the glazing. During each daylight illuminance phase, participants performed the paper-based office tasks and were then asked to respond to the visual perception questionnaire (DE) after roughly 20 minutes. The AE questions on visual perception were asked at the end of the experimental session, after all the daylight exposures, during the debriefing phase with electric lighting conditions. Indoor environmental factors were continuously recorded during the entire experimental session.

Experiments were performed only under completely clear sky conditions to guarantee constant illuminances within each experimental session. For this reason, experiments took place discontinuously from October 2017 to December 2017 and from April 2018 to June 2018. Experimental sessions completed but in which unpredicted weather conditions occurred were excluded from the analysis, resulting in the final sample size previously described (i.e., 84 participants).

2.6. Data analysis

The responses of each subjective rating were statistically evaluated with a linear mixed effects model to investigate the effects of daylight illuminance and indoor temperature

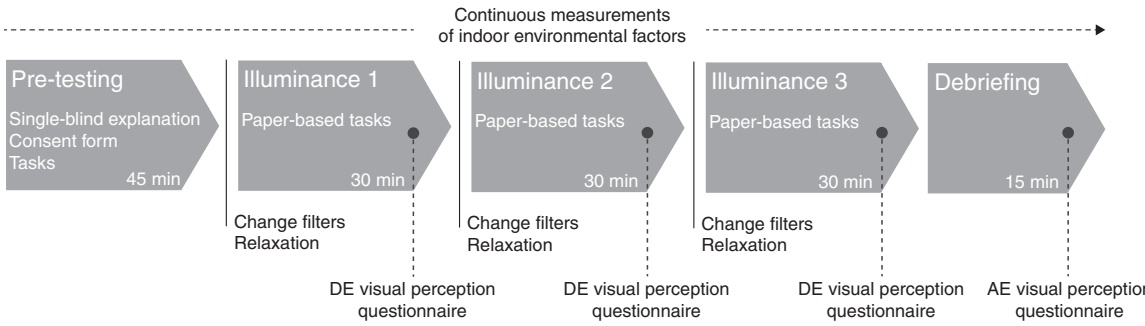


Figure 3 Experimental procedure (DE = During Exposure and AE = After Exposure).

(main effects) and their interaction (interaction effect). This type of statistical test was chosen as it considers participants as random factors of the model, making it possible to resolve the non-independence of their responses at each daylight level that violates the independence assumption of standard between-subjects ANOVA and regression model (considering the repeated design across daylight illuminances)[36].

Daylight illuminance and temperature were modelled as fixed effects, together with their interaction term ($D \times T$). Additional covariates were added to the model to consider factors that could have possibly influenced the visual perception of participants: gender[37], order of presentation of the daylight illuminances for each participant, morning and afternoon experimental sessions and the slight variations of illuminance and temperature measurements from the designed ones (calculated as the difference between the designed temperature and illuminance values and the average measured ones for each participant in each condition). Whenever the interaction term was not significant, it was excluded from the model to allow the right estimation of the main effects. Covariates were also omitted when their effects were not significant. Whenever a main effect of daylight illuminance or temperature was significant, *post-hoc* tests were performed to evaluate all possible pairwise comparisons of the three levels of each factor. Multiple testing corrections to reduce Type I error were applied with the use of the multiplicity adjustment of Tukey's honestly significant difference test[38].

In the case of significant interaction effects between daylight illuminance and temperature, additional analyses were performed at each temperature level for the investigation of the differences across daylight illuminances within one thermal condition. For this analysis, a dependent-sample Friedman test was used considering the repeated measures of responses under different daylight levels (daylight illuminance was a within-subject factor). In the case of a significant effect of daylight with the Friedman test, *post-hoc* pairwise comparisons across daylight illuminances were performed with Nemenyi *post-hoc* tests[39]. In this case, as the test was developed to account for the Type I error and is already a conservative test, no *p*-value adjustments were applied. These non-parametric tests were chosen instead of the parametric repeated measures ANOVA as responses were ordinal and not normally distributed. In the first part of the analysis (the investigation of the main effects), a parametric test was used as, to the authors' knowledge, no non-parametric alternative exists able to consider mixed design while evaluating interactions.

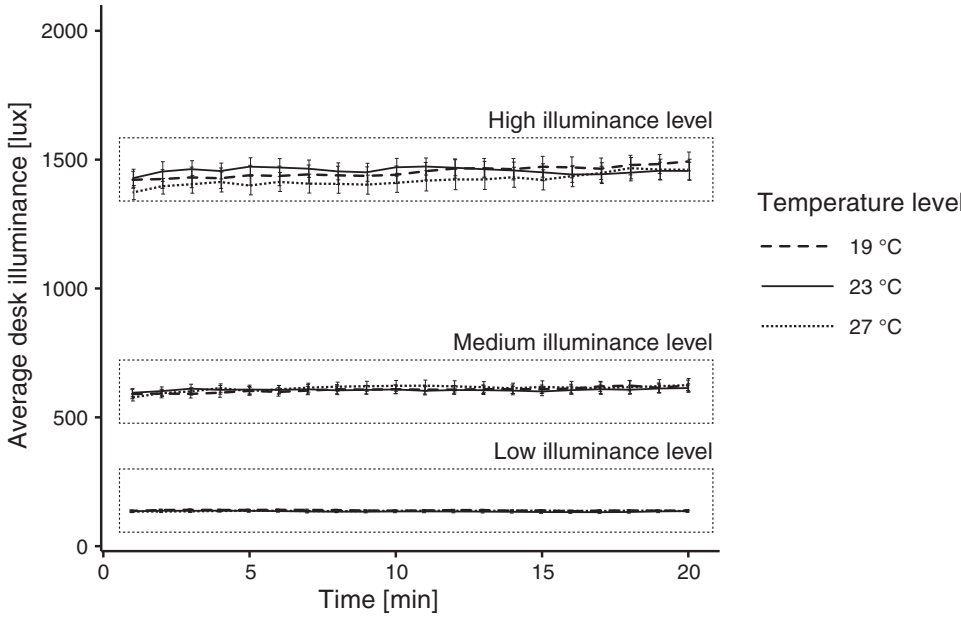


Figure 4 Average and standard error of the mean measured illuminances at each temperature level over the course of each daylight exposure, until the start of the subjective questionnaire (20 minutes).

Statistical analyses were performed using R[40] with the RStudio integrated development environment (RStudio Inc., Boston, MA). For the first part of the analysis, linear mixed models were performed with the lmerTest package[41] together with the car package for the calculation of p values[42], while *post-hoc* comparisons were calculated using the lsmeans package[43]. In the second part of the analysis, the package PMCMR was used for the Nemenyi *post-hoc* test[44]. All graphs were created with the ggplot2 package[45]. The significance level for all analyses was set at $p = 0.05$.

3. Results

3.1. Indoor environmental conditions

Indoor environmental factors were continuously recorded. Figure 4 shows the average and the standard error of the mean over time for each illuminance measured on the desk (average value of measurements of the two illuminance sensors on each desk) and for all temperature levels. Experimental days were carefully selected according to expected weather conditions to have constant illuminance values inside the test room over the entire duration of each experimental session. In addition, as already mentioned, in the case of unexpected weather variation during the day of the experiment, the entire experimental session was excluded from the analysis. Table 4 indicates the average and the standard deviation of the measured horizontal illuminance for each session and for all the daylight illuminance and temperature combinations. The corresponding correlated colour temperature (CCT) was calculated for each illuminance according to McCamy's approximation formula[46], resulting in an average of 6800 K for the low illuminance (5600 K min and 7800 K max), 6500 K for the medium illuminance (5600 K min and 7500 K

Table 4 Average horizontal illuminance measured during each daylight illuminance and temperature combination (average over the entire duration of each condition ± standard deviation).

	Daylight illuminance (lx)		
Temperature	Low	Medium	High
19 °C	139 ± 15	607 ± 88	1453 ± 193
23 °C	135 ± 24	606 ± 87	1457 ± 168
27 °C	137 ± 18	613 ± 99	1421 ± 192
Total	138 ± 19	608 ± 90	1443 ± 183

max) and 6000 K for the high illuminance (5200 K min and 7200 K max).

3.2. Visual perception ratings

Significant effects of the statistical analyses (mixed effects model analyses and of *post-hoc* tests) for all the visual perception ratings are summarised in Tables 5 and 6. The results of the main effects and of the *post-hoc* tests are not reported when the interaction term is significant. The effects of the covariates are indicated only when significant. In the following, results are discussed first in terms of unimodal main effect of daylight illuminance (considering results collapsed across temperature levels), then of cross-modal main effect of temperature (considering results collapsed across daylight illuminances) and finally in terms of cross-modal interaction effect of temperature and daylight illuminance levels (whenever interactions occurred).

3.2.1. Unimodal main effect of daylight illuminance

In this section, results are discussed for the DE visual perception ratings for lighting quantity, overall visual environment comfort and general light evaluations. Main effects of daylight for lighting pleasantness (DE7), lighting evaluation (AE1) and lighting rank (AE2) are not reported due to the presence of interactions with temperature levels (Tables 5 and 6), indicating a different effect of daylight illuminance on such ratings according to the thermal environment. Results of these ratings are reported in Section 3.2.3. Cross-modal interaction effect of temperature and daylight illuminance.

As expected, daylight illuminances had a significant effect ($p \leq 0.001$) for all the visual perception ratings on lighting quantity (DE1, DE2 and DE3). *Post-hoc* tests were therefore carried out to analyse differences across the daylight illuminances. Considering responses about visual sensation (DE1), *post-hoc* pairwise comparisons showed that low daylight level was unsurprisingly perceived as the least bright, whereas the high daylight level was the brightest, with the medium daylight evaluated in between the other two levels (DE1; Figure 5(a)). It is interesting to note that the low daylight level was only perceived on average as 'slightly low', despite being substantially lower (i.e., on average 140 lux on the work plane) than any recommended illuminance threshold[47–50]. However, it must be considered that this result refers to a short exposure time (i.e., less than 30 minutes) and does not consider any effect on work performance, alertness or any other effect after a longer exposure time. Results about visual comfort (DE2) indicated the low daylight level as the least comfortable compared to the medium and the high daylight illuminances (Figure 5(b)). Results showed that the difference between the medium and the high illuminance levels was not significant. *Post-hoc* analyses on visual preference (DE3) indicated that differences were significant for all pairwise comparisons. In particular, participants preferred a slightly brighter environment when they were exposed to low daylight levels, followed by the medium illuminance, while they did not want to change the daylight quantity when they were exposed to the high daylight levels (Figure 5(c)).

Daylight was a significant factor ($p \leq 0.001$) for the overall visual environment comfort (DE4). *Post-hoc* tests indicated that the low daylight level was significantly less

Table 5 Mixed effects model analysis and *post-hoc* test results for all the DE visual perception ratings. The main effect of daylight and temperature refers to the model without significant interactions. If interactions are significant, no main effects are reported. Significant effect with '*' $p<0.05$, '**' $p<0.01$ and '***' $p<0.001$. The symbol $^{(.)}$ indicates a marginal effect ($p<0.10$). Besides daylight illuminance and temperature factors, as well as their interaction, only the results of significant factors considered as covariates are reported. D = daylight illuminance, T = Temperature, L = Low, M = Medium and H = High.

Question [code]	Factor	Daylight or temperature mean ± standard deviation	Mixed-model result	*Post-hoc* result
Visual sensation [DE1]	D x T		F(4, 241) = 0.66, $p = 0.619$	
	D	H = 4.2 ± 0.7	F(2, 245) = 86.64, $p<0.001$***	L-H, –1.29, $p<0.001$***
		M = 3.9 ± 0.8		L-M, –0.96, $p<0.001$***
		L = 2.9 ± 0.8		M-H, –0.33, $p = 0.003$**
	T		F(2, 245) = 0.78, $p = 0.457$	
	Order of presentation		F(2, 245) = 13.22, $p<0.001$***	
Visual comfort [DE2]	D × T		F(4, 241) = 1.26, $p = 0.286$	
	D	H = 4.3 ± 0.6	F(2, 245) = 47.80, $p<0.001$***	L-H, –0.88, $p<0.001$***
		M = 4.3 ± 0.7		L-M, –0.83, $p<0.001$***
		L = 3.4 ± 0.9		M-H, –0.05, $p = 0.86$
	T		F(2, 245) = 0.11, $p = 0.894$	
	Order of presentation		F(2, 245) = 6.10, $p = 0.002$**	
Visual preference [DE3]	D × T		F(4, 241) = 0.81, $p = 0.515$	
	D	H = 3.9 ± 0.5	F(2, 245) = 86.38, $p<0.001$***	L-H, –0.91, $p<0.001$***
		M = 3.5 ± 0.6		L-M, –0.60, $p<0.001$***
		L = 2.9 ± 0.7		M-H, –0.31, p = 0.001***
	T		F(2, 245) = 0.64, $p = 0.528$	
	Order of presentation		F(2, 245) = 4.68, $p = 0.01$*	
Overall visual environment comfort [DE4]	D x T		F(4, 243) = 2.04, $p = 0.091^{(.)}$	
	D	H = 3.9 ± 0.6	F(2, 247) = 9.54, $p = 0.001$***	L-H, –0.37, $p<0.001$***
		M = 3.9 ± 0.6		L-M, –0.31, $p = 0.002$**
		L = 3.6 ± 0.8		M-H, –0.05, $p = 0.789$
	T		F(2, 247) = 0.78, $p = 0.457$	
Lighting cold–warmth association [DE5]	D × T		F(4, 242) = 1.48, $p = 0.208$	
	D	H = 4.2 ± 1.2	F(2, 246) = 12.79, $p<0.001$***	L-H, –0.75, $p<0.001$***
		M = 3.9 ± 1.2		L-M, –0.51, $p = 0.002$**
		L = 3.4 ± 1.2		M-H, –0.24, $p = 0.260$
	T	19 °C = 3.5 ± 1.3	F(2, 246) = 7.17, p = 0.001**	19-23 °C, –0.27, $p = 0.422$
		23 °C = 3.8 ± 1.1		19-27 °C, –0.81, $p = 0.001$**
		27 °C = 4.3 ± 1.1		23-27 °C, –0.53, $p = 0.041$*
	Gender		F(1, 246) = 4.23, $p = 0.04$*	W-M, –0.36, $p = 0.042$*
Lighting pleasantness [DE7]	D × T		F(4, 241) = 3.08, $p = 0.017$*	
	D		–	
	T		–	
	Order of presentation		F(2, 245) = 5.81, $p = 0.003$**	

Table 6 Mixed effects model analysis and *post-hoc* test results for all the AE visual perception ratings. The main effect of daylight and temperature refers to the model without interactions. If interactions are significant, no main effects are reported. Besides daylight illuminance and temperature factors, as well as their interaction, only the results of significant factors considered as covariates are reported. Significant effect with '*' $p<0.05$, '**' $p<0.01$ and '***' $p<0.001$. D = Daylight illuminance and T = Temperature.

Question [code]	Factor	Mixed-model result
Lighting evaluation [AE1]	D × T	$F(4, 243)=2.51$, $p=0.043$*
	D	–
	T	–
Lighting rank [AE2]	D × T	$F(4, 241)=2.49$, $p=0.043$*
	D	–
	T	–
	Order of presentation	$F(4, 245)=3.08$, $p=0.047$*

comfortable than both the medium and the high daylight levels. However, as visible in Figure 5(d) and from the statistical analysis (Table 5), a slight interaction occurred between temperature and daylight levels ($p<0.10$). For this reason, further investigations are reported in the section on the combined effect of daylight and temperature. It is important to note the different results between responses on overall visual environment comfort (DE4) and on visual comfort (DE2), shown in Figure 5: the medium and the high daylight conditions were rated generally more comfortable when participants estimated the light quantity on the desk (DE2), compared to their responses about the overall visual environment comfort (DE4). Moreover, the overall visual environment comfort (DE4) under the low illuminance condition increased with temperature, while such an increment was not visible for the comfort evaluation of the light quantity (DE2). Such differences indicate that the participants do not consider daylight quantity (i.e., brightness) as the only factor affecting their overall visual comfort perception of the visual environment.

Daylight illuminance was a significant factor also for the cold–warmth association (DE5). *Post-hoc* tests indicated that the perception of the low daylight illuminance was colder than that of both the medium and the high daylight illuminance (Figure 6(a)), possibly due to the fact that its average CCT was higher (i.e., 'colder') than that of the other two levels or because a dim environment can be associated with colder spaces (e.g., caves or churches).

3.2.2. Cross-modal main effect of temperature

Temperature resulted as a significant factor for only one visual perception rating, namely the cold–warmth association of the lighting (DE5 – Figure 6(a)). *Post-hoc* tests revealed that daylight was associated more often to a warmth adjective in warm temperatures compared to colder ones. Specifically, the responses under the highest temperature level (27 °C) were substantially different in comparison with those under both the neutral (23 °C) and the cold (19 °C) thermal environments. Despite the fact that the main effect of temperature indicated that results were independent from the daylight illuminance, responses under the medium daylight condition were the ones less affected by temperature (Figure 6(a)). This is confirmed by the analysis of the simple effects of temperature for each daylight level performed with a non-parametric Kruskal–Wallis test to account for the independence of responses at each temperature level. Results of these tests indicate that significant results across temperature levels were found for the low ($p=0.02$) and

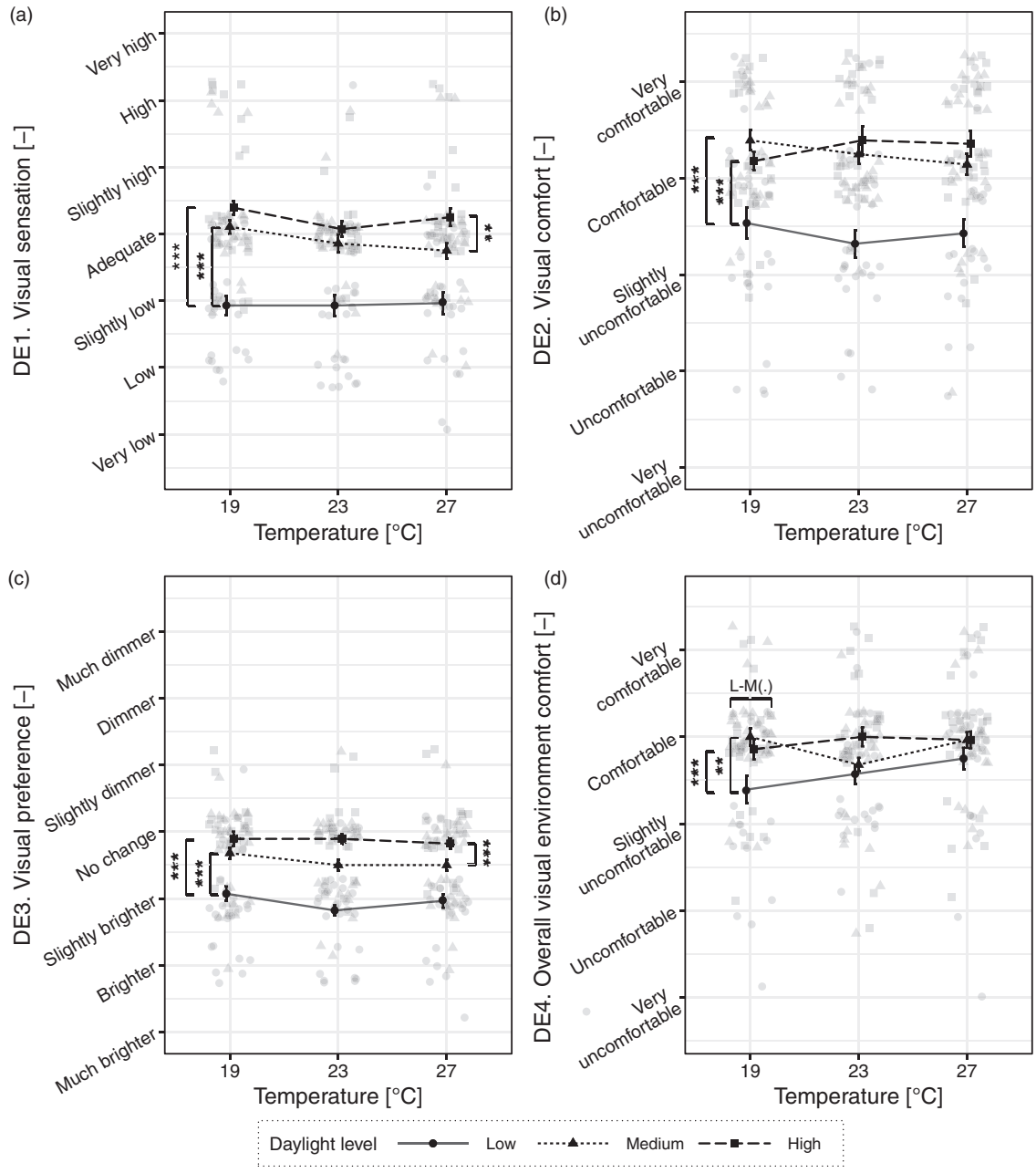


Figure 5 Average and standard error of the mean of the DE visual perception ratings for each daylight illuminance and temperature combination. (a) Visual sensation (DE1); (b) visual comfort (DE2); (c) visual preference (DE3) and (d) overall visual environment comfort (DE4). The grey points in the background represent the raw data. Unimodal main effect of daylight levels indicated on the side with vertical lines, cross-modal main effect of temperature indicated on the top with horizontal lines and significant difference between two levels of daylight (L = low, M = medium and H = high) reported at a specific temperature in the case of significant interactions. Significance levels indicated with '.' $p < 0.10$, '*' $p < 0.05$, '**' $p < 0.01$ and '***' $p < 0.001$.

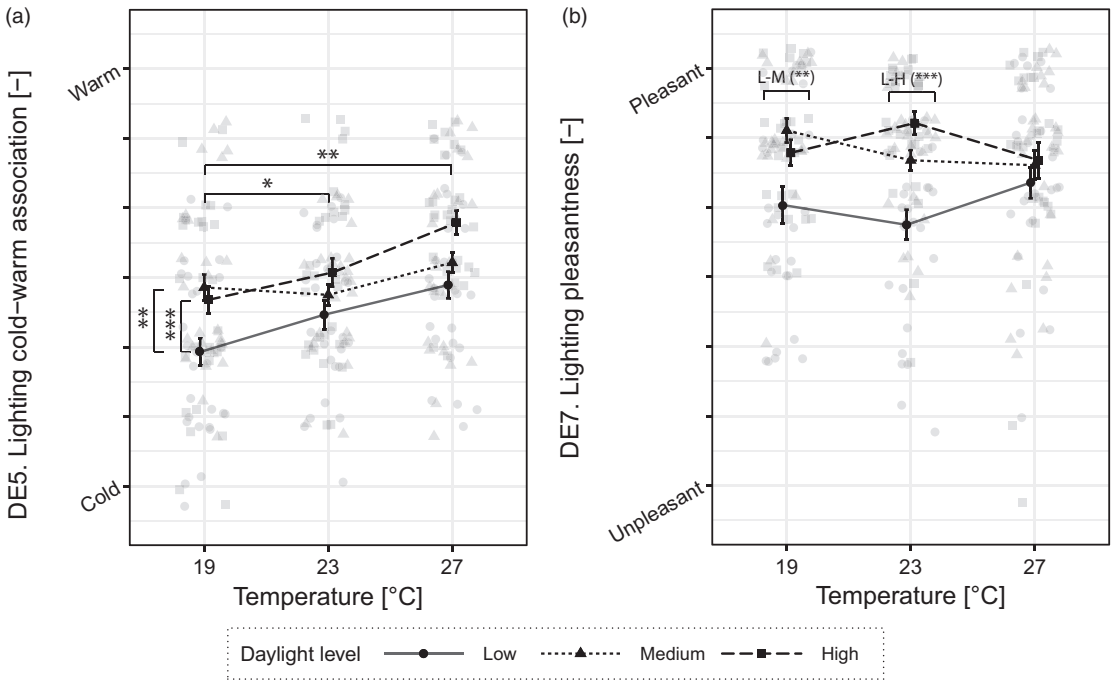


Figure 6 Average and standard error of the mean of the DE visual perception ratings for each daylight illuminance and temperature combination. (a) Lighting cold–warmth association (DE5) and (b) lighting pleasantness (DE7). The grey points in the background represent the raw data. Unimodal main effect of daylight levels indicated on the side with vertical lines, cross-modal main effect of temperature indicated on the top with horizontal lines and significant difference between two levels of daylight (L = low, M = medium and H = high) reported at a specific temperature in the case of significant interactions. Significance levels indicated with '.' $p<0.10$, '*' $p<0.05$, '**' $p<0.01$ and '***' $p<0.001$.

the high daylight conditions ($p=0.008$), but not for the medium one. Hence, it can be concluded that the cross-modal main effect of temperature on the daylight cold–warmth association was stronger for the extreme daylight conditions (i.e., low and high).

3.2.3. *Cross-modal interaction effect of temperature and daylight illuminance*

This section presents the results of the statistical analyses performed for the visual perception ratings for which a significant (or marginal) interaction between daylight and temperature was reported in the main analysis (Tables 5 and 6), indicating that the effect of temperature on such visual perceptions depended on the daylight illuminance.

With reference to the DE visual perception ratings, a significant interaction effect is reported only for the lighting pleasantness (DE7; Table 5). Analyses conducted at each temperature level showed that differences across daylight levels were significant at 19 °C ($\chi^2(2)=10.64$, $p=0.004$) and at 23 °C ($\chi^2(2)=27.57$, $p<0.001$), but not at 27 °C. *Post-hoc* tests revealed that the low daylight exposure was considered less pleasant than the high daylight exposure at 23 °C ($p<0.001$), and less pleasant than the medium daylight exposure at 19 °C ($p=0.01$; Figure 6(b)). This result indicates an interaction between daylight illuminance and temperature levels, with the low daylight level generally less pleasant than the medium and the high daylight level but only at 19 °C

and at 23 °C. In the warm thermal condition (27 °C), all daylight levels were considered equally pleasant. This latter result suggests that low illuminances are more pleasant at higher temperatures.

For the overall visual environment comfort (DE4), the interaction term had a marginally significant effect ($p < 0.10$; Table 5). Analyses conducted at each temperature level report results similar to the ones for the lighting pleasantness (DE7), described above. In particular, differences across daylight levels were significant at 19 °C ($\chi^2(2) = 10.61$, $p = 0.004$) and at 23 °C ($\chi^2(2) = 7.50$, $p = 0.02$), but not at 27 °C. Nevertheless, *post-hoc* tests revealed that only at 19 °C and only between the medium and the low conditions were the differences in ratings marginally significant ($p = 0.051$).

Analyses of the two AE visual perception ratings showed a significant effect of the interaction term (Table 6). Results of the Friedman tests conducted at each temperature level for the lighting evaluation (AE1) indicated that only at 23 °C were differences across daylight levels significant ($\chi^2(2) = 17.63$, $p < 0.001$), while they were only marginally significant at 19 °C ($\chi^2(2) = 4.70$, $p = 0.09$). *Post-hoc* tests comparing daylight levels at 23 °C revealed that the low daylight condition was significantly less pleasant than both the medium ($p = 0.006$) and the high ($p < 0.001$) conditions (Figure 7(a)). Identical results were found for the lighting rank responses (AE2; Figure 7(b)): only a marginally significant difference across daylight levels was found at 19 °C ($\chi^2(2) = 5.42$, $p = 0.06$), whereas a significant difference occurred at 23 °C ($\chi^2(2) = 20.21$, $p < 0.001$), with the low daylight condition being less chosen compared to the medium ($p = 0.002$) and the high ($p < 0.001$) daylight conditions.

3.2.4. Confounding factors effects

The confounding factors having a significant effect on the visual perception ratings are indicated in Tables 5 and 6. In general, the order of presentation of the daylight illuminances (counterbalanced across participants and temperature levels) was the only significant factor, but just for a few ratings (DE1, DE2, DE3 and DE7). The only exception is the gender of participants, which significantly affected the apparent warmth of daylight, with women associating the light to a cold adjective more often than men. This is an interesting finding that might be linked to the general colder thermal perception of women in indoor environments reported in some controlled experiments and field studies[51].

4. Discussion

As introduced previously, this is the first experiment evaluating the simultaneous effect of different illuminances (below discomfort thresholds) and temperatures on visual perception, with the use of daylight as the only source of light. For this reason, the following discussion refers to investigations considering electric light only[24,25,29] or investigating daylight but without evaluating different controlled and constant illuminances below discomfort thresholds[26,30]. The discussion is divided into two sections, analysing the cross-modal main effect of temperature (as well as its interaction with daylight illuminances) on visual perception and the unimodal effect of daylight illuminances on visual perception.

4.1. Cross-modal effect of temperature

Our study with the use of daylight confirmed results about the cold–warmth association evaluation of light reported in te Kulve *et al.*[24] with the use of electric light: there exists a cross-modal effect of temperature on visual perception. However, our results agree only for the high illuminance under which, in both studies, an increase of the warmth association with an increase of temperature is reported. On the other hand, the studies

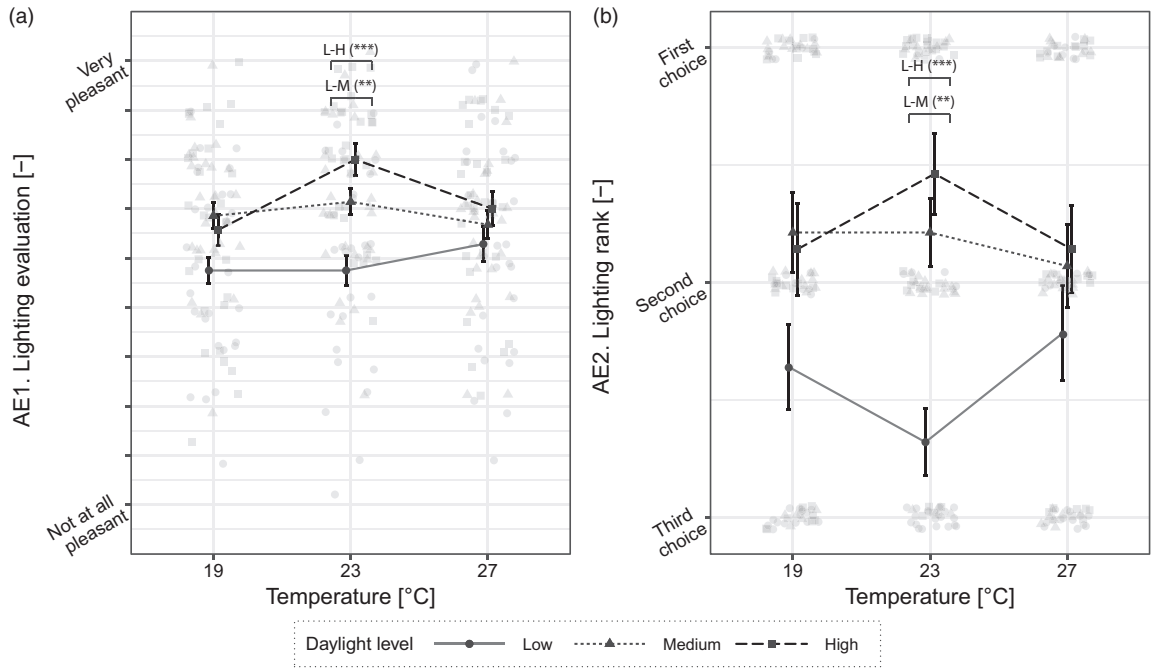


Figure 7 Average and standard error of the mean of the AE visual perception for each daylight illuminance and temperature combination. (a) Lighting evaluation (AE1) and (b) lighting rank (AE2). The grey points in the background represent the raw data. Unimodal main effect of daylight levels indicated on the side with vertical lines, cross-modal main effect of temperature indicated on the top with horizontal lines and significant difference between two levels of daylight (L = low, M = medium and H = high) reported at a particular temperature in case of significant interactions. Significance levels indicated with '.' $p<0.10$, '*' $p<0.05$, '**' $p<0.01$ and '***' $p<0.001$.

disagree for the low illuminance: our study shows an increase of the warmth association with an increase of temperature for the high illuminance, whereas te Kulve *et al*. found the opposite behaviour, i.e., a 'warmer' association with decreasing temperature. This contrasting result might be explained by the very low light level used in the study with electric light ($\sim$ 5 lux), which is unusual in buildings, and which is lower than that used in the current investigation ($\sim$ 140 lux). To further support the results of the present study, an increasing apparent warmth with temperature was also found when electric light with different CCTs at low illuminance ($\sim$ 55 lux) was investigated, as reported in the second experiment described by te Kulve *et al*.[24]. An increasing apparent warmth association with temperature was also found in another study[30], investigating daylight transmitted through coloured glazing at different indoor temperatures. Overall, it can be concluded that, in agreement with previous studies on electric lighting or with 'coloured daylight', temperature affected the apparent warmth association of different quantities of daylight, with daylight associated more often with a warm adjective in a warm thermal condition compared to a colder one. This result indicates that visual perception of people can be unconsciously affected by other indoor factors, besides the visual ones. In this case, even though the question was explicitly referring to the evaluation of daylight, indoor temperature indirectly influenced the responses of participants.

Interestingly, statistical results of this study indicated that the apparent warmth was also affected by the gender of participants, with women associating all the daylight illuminances to a cold adjective more often than men. Past studies investigating gender effect on visual perception either did not study the lighting cold–warmth association[25,26,29] or did not report a gender effect on other types of visual evaluations[30]. However, in other studies, gender effects on visual perceptions have been reported in terms of visual sensation and pleasantness[26] or visual comfort[29]. Additional investigations are needed to further the understanding of the gender effect on visual perception.

Contrasting conclusions with those reported in studies on electric light can be found for other visual perception evaluations, in terms of brightness evaluation (i.e., visual sensation) and visual comfort. First of all, contrary to what was reported in other studies on electric light[24–26], temperature did not influence the apparent brightness evaluation of daylight levels, as an equal apparent brightness was reported at all temperatures. On the other hand, in agreement with results of the current investigation, no cross-modal effect of temperature on brightness evaluation was found when daylight was used as the source of light in past investigations, whether alone, in combination with electric light or transmitted through coloured glazing[26,30]. In addition, the findings of the present study suggest that the independence of brightness evaluation of daylight from the thermal environment holds true for all the illuminances investigated, a conclusion not reached in previous studies as illuminances of daylight were not strictly controlled but only recorded[26,30]. The differences observed between the results of studies on electric light and those of studies on daylight could be explained in terms of absolute rating values, which appeared more extreme in the experiments with the electric light[24,25] in comparison with the current investigation on daylight. In particular, a lower apparent brightness evaluation was reported under electric light than under daylight, when the dim condition was evaluated. Similarly, for the high illuminance, a higher apparent brightness was indicated under electric light than under daylight (even if the illuminance was lower under electric light – $\sim$ 1000 lux – than under daylight $\sim$ 1440 lux). The cross-modal effect of temperature on brightness evaluation might have been more evident under electric light in past studies due to the more extreme perceived brightness evaluations.

Another difference from results referring to electric light[23] lies in the 'visual comfort' evaluation. In the present study, no cross-modal effect of temperature on visual comfort evaluation was detected. This is in contrast with results on electric light reporting that the low illuminance was perceived as less comfortable in the warm thermal condition compared to the neutral one[24].

Results in terms of lighting pleasantness, evaluation and rank cannot be compared with investigations on electric light, as these subjective ratings were not investigated in previous studies. Results of the current investigation indicated that interactions between temperature and daylight illuminances occurred for the aforementioned visual evaluations, during both the DE and AE phases. In particular, the high illuminance was more pleasant, better evaluated and chosen more often (ranked) than the low one only at 23 °C (considered a thermally comfortable condition), while all illuminances were rated equally in the warm thermal condition (27 °C). This result was observed as the evaluation of the low illuminance condition increased with increasing temperatures and that of the higher illuminance conditions decreased.

In conclusion, cross-modal effects of temperature were observed for lighting

pleasantness, evaluation and rank, according to the daylight illuminance, indicating interaction effects between the two factors. This result suggests that thermal conditions can affect the perception of the light quantity in a space, with high illuminances preferred more than lower ones only in a thermally comfortable environment and with low illuminance levels better evaluated in warm environments than in cold ones.

4.2. Unimodal effect of daylight illuminances

Besides the described results about the cross-modal effect of temperature on visual perception, another finding of interest is the difference in visual perception evaluations according to daylight levels (i.e., a unimodal effect).

From the comparison of the results at different daylight illuminances, it was found that the largest differences in visual perception evaluations were reported between the low and the high conditions, as well as between the low and the medium conditions. Differences between the medium and the high illuminance exposures were substantial only in terms of visual sensation and preference. The other types of visual evaluations did not statistically differ under the medium and the high illuminance exposures, especially in terms of comfort (related to both lighting quantity and overall visual environment). This means that an increase in illuminance, once already above a comfortable threshold (in this case $\sim$ 610 lx), would not result in a visual comfort increase. However, it must be noted that such responses occurred under daylight primarily diffused by the north openings and that people's visual comfort could differ in case of evaluation of sunlight directly incident on their desk.

Another interesting outcome from a comparison of responses of this study with those reported in previous investigations on electric light, regards visual sensation evaluation (i.e., brightness evaluation): similar high illuminances were evaluated as being too bright under electric light conditions in the past studies[24,25] and just above adequate under daylight in the present study. These results were observed even though the illuminances were higher for daylight in the present study ($\sim$ 1440 lx) than for electric light in past studies ($\sim$ 1000 lx). They confirm the results of another study evaluating daylight illuminances[52] in which participants indicated that the illuminance at the work plane was too high only above 4500 lx and in which participants decided to work on very high illuminances (around 3000 lx on average), by modifying the position of window blinds. We hypothesise that higher illuminances could be more tolerated under daylight compared to electric light. As a possible explanation, high daylight illuminances (still below discomfort thresholds) could be more tolerable than comparable illuminances under electric light because people are used to high illuminances outdoors. It must be noted, however, that the different results between the present and past studies can also be determined by other factors than the type of light source, such as the type of task performed by participants (e.g., use of electronic devices) or the experimental design (e.g., exposure time and light distribution). Differences in terms of subjective ratings evaluated and the stimuli intensities used make the comparison across studies even more difficult. In conclusion, considering that the present experiment was not designed to compare brightness perceptions between electric light and daylight illuminances (e.g., due to differences in light distribution: in this experiment, participants were primarily exposed to diffuse light from the North window, which is not the same as that usually experienced from electric light – i.e., direct illumination), further investigations explicitly designed for this purpose should be conducted to test our hypothesis and to better understand the differences between visual

perception evaluations under daylight and electric light.

4. Conclusions

This study described the first investigation on the combined effect of temperature (19 °C, 23 °C and 27 °C) and working plane horizontal illuminance (low: ~140 lx, medium: ~610 lx and high: ~1440 lx) on visual perception, with the use of daylight as the only source of light. Experiments were conducted in an office-like test room and involved 84 participants. Statistical analyses were performed to investigate the cross-modal effect of temperature on visual perception, as well as the unimodal influence of different daylight illuminances on visual perception.

Results showed that the cross-modal effects of temperature on visual perception occurred as specific evaluations of daylight were affected by the thermal environment people were exposed to. In particular, the apparent warmth of daylight increased with temperature, especially for the low and the high illuminances, with participants more often associating daylight with a warm adjective at higher temperatures compared to colder ones. Moreover, interactions between daylight illuminances and temperature influenced the result of lighting pleasantness, evaluation and rank: the high daylight illuminance condition was more pleasant, better evaluated and chosen more often than the low one only at 23 °C, considered as a comfortable thermal environment. All illuminances were rated equally in the warm thermal condition (27 °C) as the evaluation of the low illuminance increased with temperature and that of the higher illuminances decreased. Contrary to past investigations on electric light, temperature did not influence visual sensation (brightness perception) and visual comfort.

Analyses on the unimodal effect of daylight illuminances on visual perception indicated that no significant differences in visual perception evaluations were found between the medium and the high illuminances, especially in terms of visual comfort. This result indicates that daylight illuminances above an already comfortable value do not lead to an increase of visual comfort. Comparing results with those of past studies, it was also shown that more extreme brightness evaluations for both the low and the high illuminances were reported under electric light compared to daylight. As high illuminances were perceived brighter under electric light compared to similar values under daylight, it was hypothesised that high illuminances might be more tolerated in a naturally-lit environment compared to a space illuminated with electric light.

To summarise the most relevant results about the cross-modal and unimodal effects of daylight and temperature on visual perception, it can be concluded that high daylight illuminances below discomfort thresholds (i.e., non-glaring) do not result in higher visual comfort above an already comfortable value (in this case, above 610 lx) and are better evaluated than low illuminances only in thermally comfortable conditions. Such findings are meaningful for both research and practice for two main reasons: first, they highlight that thermal conditions should be considered in investigations on visual perception of daylight as well as in standards and guidelines applied in practice for building design and operation to guarantee a pleasant visual environment. Second, they advance the understanding of occupants' visual perceptions in naturally-lit spaces, which might differ from those occurring under electric lighting.

Acknowledgements

The authors would like to thank the researchers and professors at the International Centre for Indoor Environment and Energy at the Technical University of Denmark (DTU, Copenhagen) for their valuable support. In

particular, many thanks to Prof. Toftum, Prof. Wargocki and Prof. Olesen for their advice and support on the experimental design and the equipment used in the experiment. All participants are thanked for taking part in this study.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The authors disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: by the Ecole Polytechnique Fédérale de Lausanne (EPFL).

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